Effects of Whole Grain Intake, Compared to Refined Grain, on Appetite and Energy Intake: A Systematic Review and Meta-Analysis

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Online Supplementary Material

Supplemental Table 1. Search terms used to identify relevant RCT publication for the systematic review and meta-analysis of WG and appetite.

Terms used to identify all studies	Additional terms used to limit identified studies to RCTs
Key search terms for WG "whole grain*" OR wholegrain* OR whole-grain* OR "whole meal" OR wholemeal OR whole-meal OR "whole wheat" OR wholewheat OR whole-wheat OR "brown rice" OR "wild rice" OR "purple rice" OR "black rice" OR "red rice" OR "whole rice" OR "whole barley" OR "hulled barley" OR "hull-less barley" OR maize OR corn OR "whole corn" OR popcorn OR "whole rye" OR "whole oat" OR "oat" OR "oats" OR "oatmeal*" OR millet OR fonio OR sorghum OR milo OR teff OR triticale OR amaranth OR buckwheat OR quinoa OR kaniwa OR canihua OR spelt OR emmer OR faro OR farro OR einkorn OR kamut OR durum OR bulgur OR freekeh **Key search terms for appetite and satiety** appetite OR satiety OR hunger OR fullness OR desire to eat OR "prospective food consumption" OR "energy intake*" OR "caloric intake*" OR "dietary intake*" OR "food intake*" OR appetite hormone* OR "gut hormone*" OR amylin OR cholecystokinin OR CCK OR "corticotropin releasing factor" OR CRF OR dopamine OR ghrelin OR "glucose-dependent insulinotropic polypeptide" OR GIP OR "glucagon-like peptide" 1 OR GLP-1 OR glucagon OR insulin OR leptin OR oxyntomodulin OR OXM OR "pancreatic polypeptide" OR PP OR "peptide YY" OR PYY OR serotonin OR "transit time" OR "gastrointestinal emptying time" OR "gastric emptying time"	randomized controlled trial OR controlled clinical trial OR RCT OR random OR placebo OR control OR clinical trial OR trial OR clinical research stud* OR random allocation OR double-blind method OR single-blind method OR comparative stud* OR evaluation stud* OR cross-over stud* OR latin square OR intervention stud* OR dietary intervention*

Supplemental Table 3. Risk of bias assessment

Author	Outcome	Randomization Process	Deviation from intended interventions	Missing outcome data	Measurement of the Outcome	Selection of the reported results	Overall Bias
Forsberg 2014 (21)	Subjective appetite	Some concerns	Some concerns	Low	Low	Low	Some concerns
Johansson 2015 (37)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Gonzalez-Anton 2015 (36)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Geliebter 2015 (35)	Subjective appetite	High	Low	Low	Low	Low	High
Sandberg 2016 (34)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Lee 2017 (30)	Subjective appetite	Low	Low	Low	Low	Low	Low
Cioffi 2016 (32)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Cioffi 2016 (33)	Subjective appetite	Some concerns	Low	Low	Low	Low	Some concerns
Granfeldt 1994 (46)	Subjective appetite	Some concerns	Low	Low	High	Low	High
Hamad, 2018 (47)	subjective appetite	Some concerns	Low	Low	Low	High	High
Sandberg 2017 (31)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Hlebowicz 2007 (45)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Solah 2007 (22)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Kristensen 2010 (43)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Rosen 2009 (44)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Rosen 2011 (40)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Rosen 2011 (41)	Subjective appetite	Some concerns	Low	Low	Low	High	High
Rosen 2011 (42)	Subjective appetite	Some concerns	Low	Low	Low	High	High
Johansson 2013 (39)	Subjective appetite	Some concerns*	Low	Low	Low	Low	Low
Hartvigson 2014 (38)	Subjective appetite	Low	Low	Low	Low	Low	Low
Costabile 2018 (29)	Subjective appetite	Low	Low	Low	Low	Low	Low
Wolever 2019 (28)	Subjective appetite	Low	Low	Low	Low	Low	Low
Kristensen 2010 (43)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Rosen 2011 (40)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Johansson 2015 (37)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Hartvigson 2014 (38)	Energy intake	Low	Low	Low	Low	Low	Low

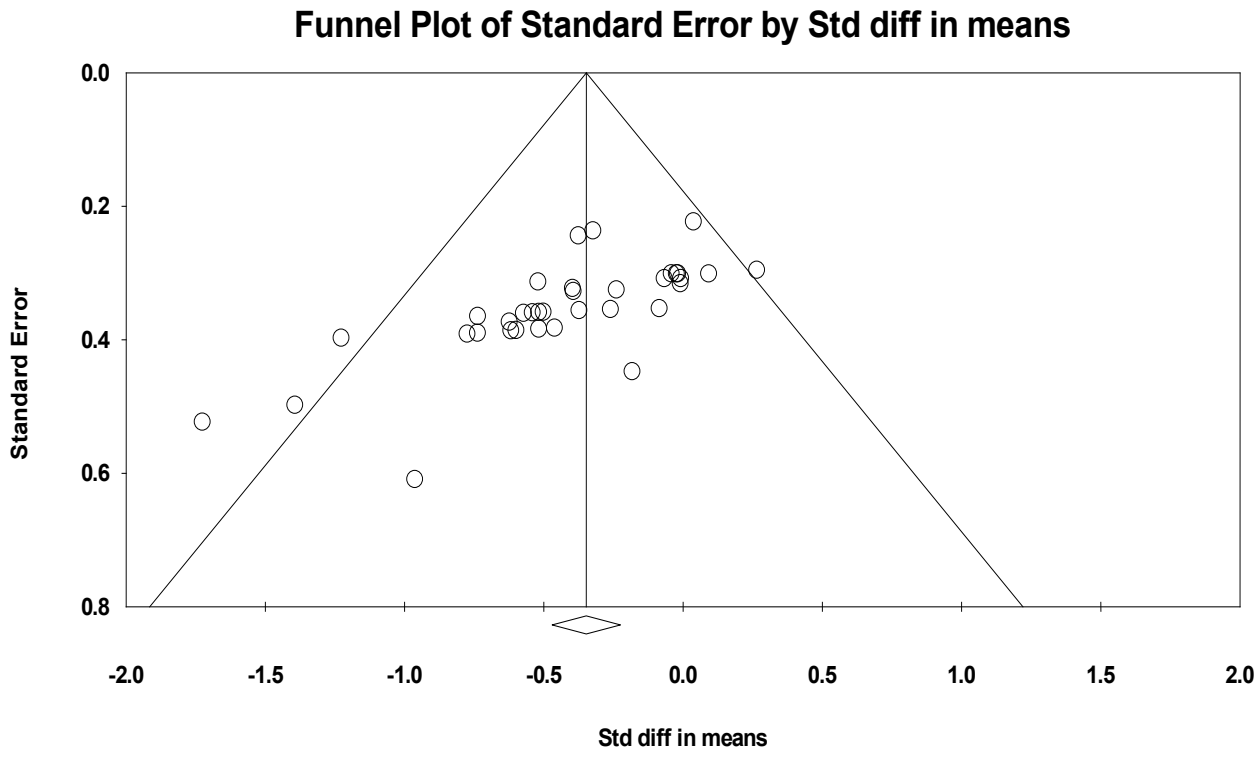
Author	Outcome	Randomization Process	Deviation from intended interventions	Missing outcome data	Measurement of the Outcome	Selection of the reported results	Overall Bias
Gonzalez-Anton 2015 (36)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Cioffi 2016 (32)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Sandberg 2017 (31)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Costabile 2018 (29)	Energy intake	Low	Low	Low	Low	Low	Low
Berti 2005 (56)	Energy intake	Some concerns	Some concerns	Some concerns	Low	Low	Some concerns
Isaksson 2008 (55)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Schroeder 2009 (54)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Anderson 2010 (53)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Zafar 2011 (52)	Energy intake	Some concerns	Some concerns	Low	Low	Low	Some concerns
Isaksson 2012 (51)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Ibrugger 2014 (49)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Luhovyy 2014 (50)	Energy intake	Some concerns*	Low	Low	Low	Low	Low
Roager 2019 (13)	Energy intake	Low	Low	Low	Low	Low	Low
Suhr 2017 (48)	Energy intake	Some concerns*	Low	Low	Low	Low	Low

*studies did not report information on allocation concealment, but all other aspects were low so overall bias was ranked low

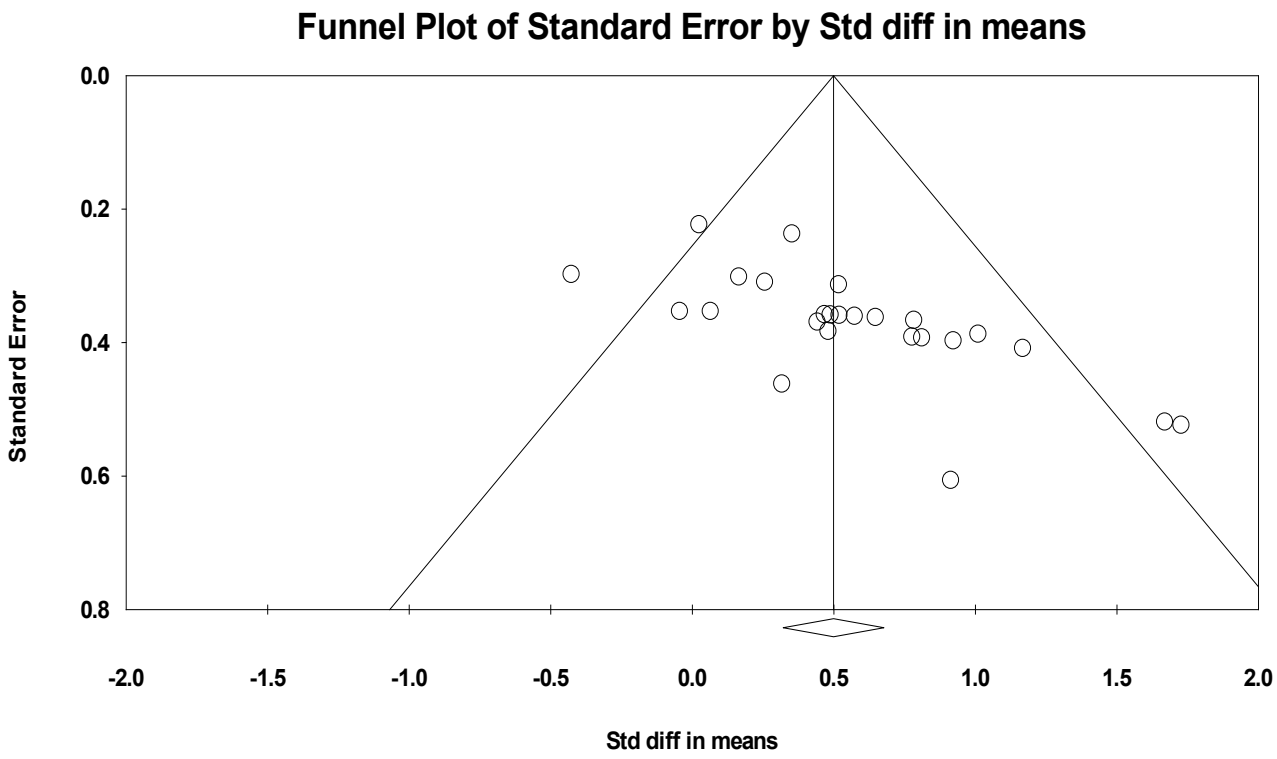
Supplemental Table 2. Sensitivity Analyses for subjective appetite measures

Outcome	Number of comparisons/studies	Subjects (WG/control)	Effect estimate SMD (95% CI)[1]	I^2 (%)	p-value[1]
Hunger					
≥ 180 min	31/16	1181 (589/592)	-0.31 (-0.42, -0.19)	0.00	<0.001
< 180 min	6/3	182 (91/91)	-0.54 (-1.01, -0.07)	58.30	0.025
Fullness					
≥ 180 min	22/11	794 (397/397)	0.42 (0.26, 0.57)	19.84	<0.001
< 180 min	5/2	144 (72/72)	0.77 (0.22, 1.33)	60.21	0.007
Desire to Eat					
≥ 180 min	21/9	796 (398/398)	-0.30 (-0.44, -0.17)	0.00	<0.001
< 180 min	6/3	181 (90/91)	-0.47 (-.96, 0.022)	62.29	0.061
Satiety					
≥ 180 min	22/11	647 (320/327)	0.33 (0.17, 0.48)	0.00	<0.001
< 180 min	2/2	62 (31/31)	0.35 (-0.15, 0.85)	0.00	0.172

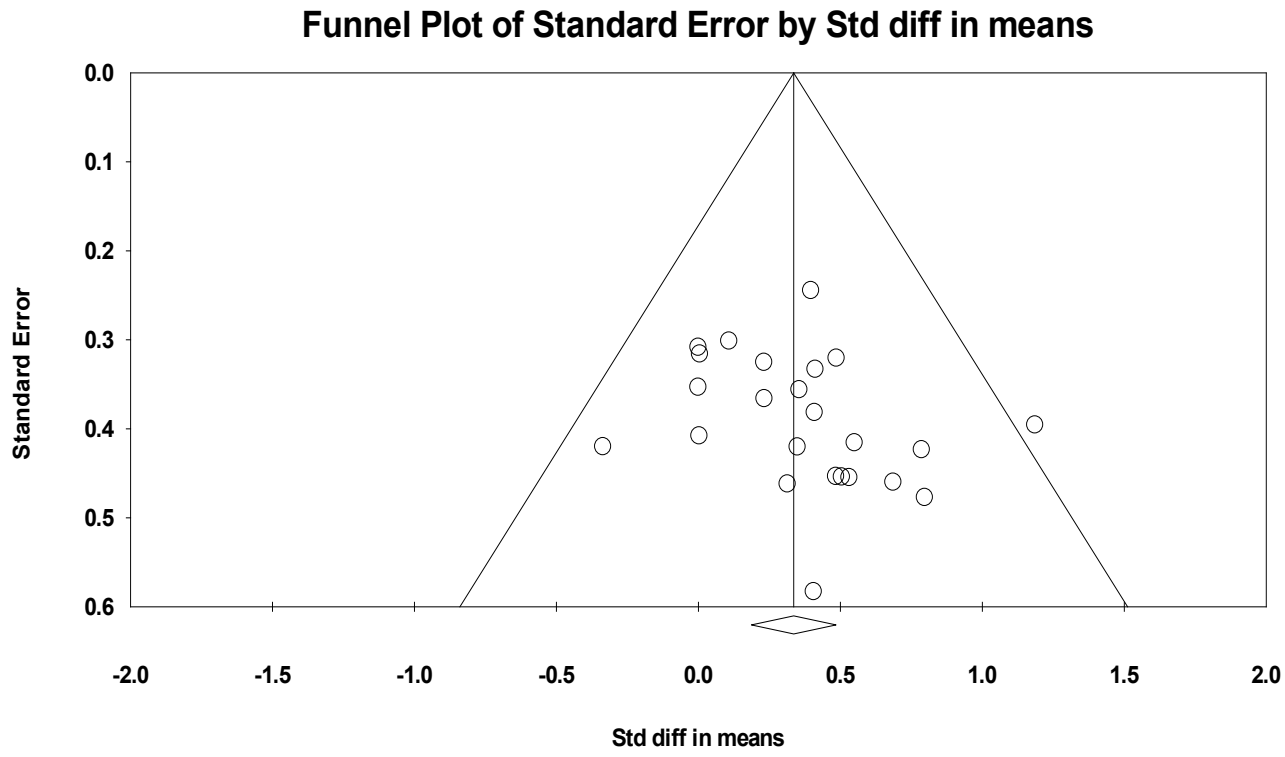
Supplemental Figure 1. Funnel plot for hunger AUC



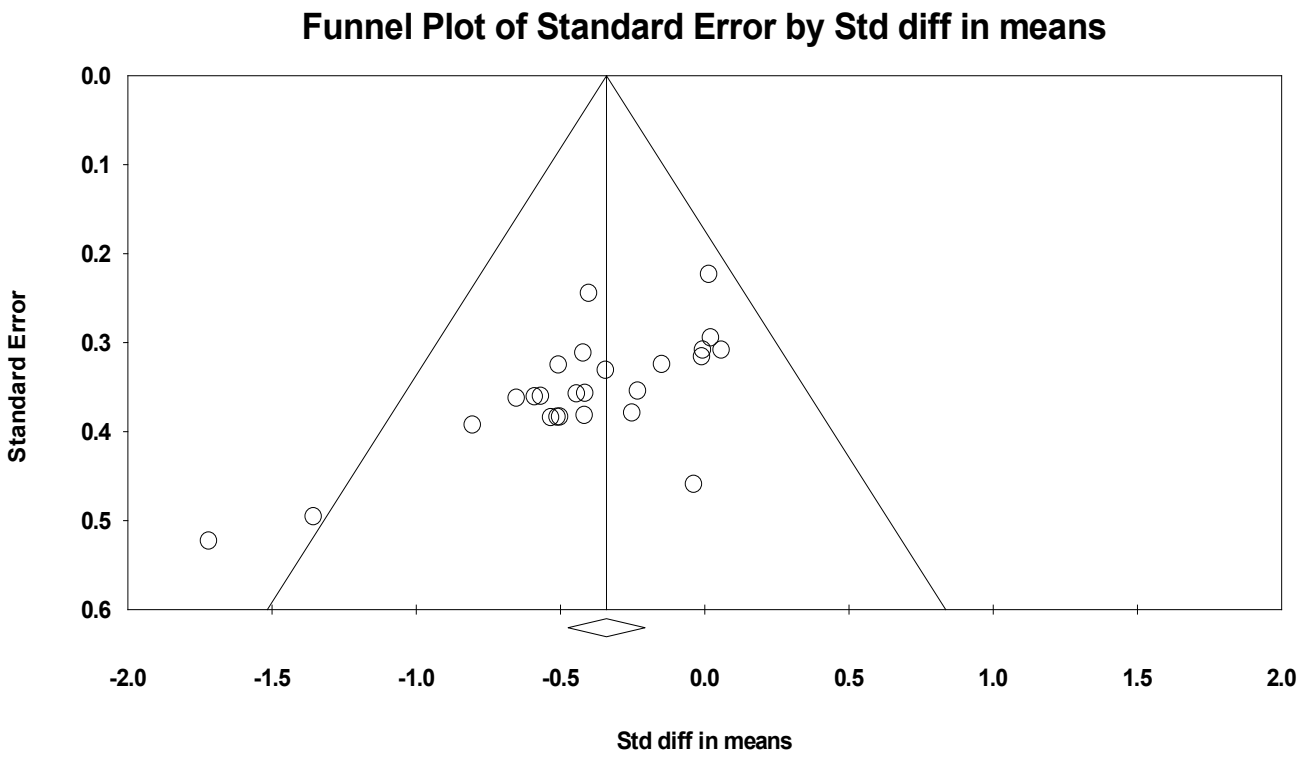
Supplemental Figure 2. Funnel plot for fullness AUC



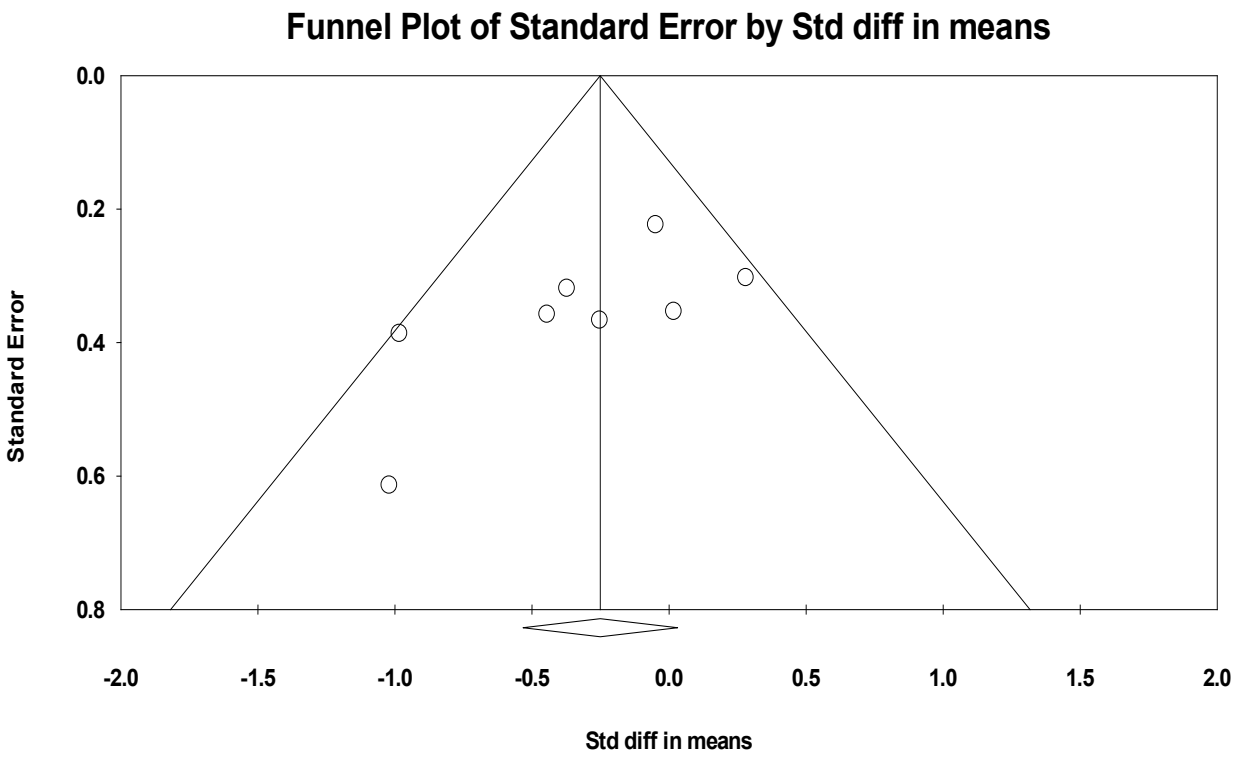
Supplemental Figure 3. Funnel plot for satiety AUC



Supplemental Figure 4. Funnel plot for desire to eat AUC



Supplemental Figure 5. Funnel plot for prospective consumption AUC



Supplemental Figure 6. Funnel plot for energy intake

